PROJECT PRESENTATION & DOCUMENTATION

GOLDCARE

A Modern, Responsive Jewellery Repair Booking, Real-Time Tracking & UPI Payment Escrow Management Platform

Student / Developer:	Vaibhav Mote
Project Domain:	Full-Stack Frontend & Web Engineering
Core Technologies:	React.js 19, Vite 8, Tailwind CSS v4
Routing & Icons:	React Router DOM (HashRouter), Lucide Icons
Data Architecture:	Client-Side Persistence (LocalStorage API)
Deployment:	GitHub Pages (CI/CD GitHub Actions)
Live Project URL:	https://vaibhav-mote.github.io/GoldCare/
GitHub Repository:	https://github.com/Vaibhav-Mote/GoldCare

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1. Executive Summary & Abstract

The jewellery repair industry in India and worldwide relies predominantly on unorganized, paper-based workflows. Customers face significant friction points including lack of transparent pricing, absence of digital repair progress tracking, insecure cash advances, and difficulty discovering verified artisanal goldsmiths.

GoldCare is an end-to-end, single-page application (SPA) created to digitize and elevate the jewellery repair ecosystem. Built with a luxurious aesthetic inspired by high-end jewellery houses (gold, cream, and deep espresso tones), GoldCare provides customers with instant access to catalogued repair services, verified local workshops, an interactive 3-step repair booking engine, a real-time scannable **UPI QR code payment gateway**, and milestone tracking.

Simultaneously, GoldCare equips shop owners and platform administrators with a dedicated, protected management back-office to oversee partner shops, modify repair statuses, maintain customer records, export CSV business analytics, and track payments without requiring any external server infrastructure.

> **Architecture Philosophy: 100% Serverless & Zero-Backend**
>
> The entire application runs as a high-performance static web application hosted freely on GitHub Pages. All state mutations, shop profiles, repair bookings, and administrative settings persist reliably in the browser's **LocalStorage API**, making it lightweight, zero-maintenance, and cost-free to operate.

2. Problem Statement & Key Objectives

Traditional Industry Challenges

- Loss of paper receipts and warranty claims.
- Uncertainty regarding repair milestones and completion dates.
- No upfront cost estimates or transparent advance handling.
- Inability for artisans to showcase verified specializations.

GoldCare Solutions

- Unique Tracking ID (e.g., `GC-2026-0001`) with digital receipts.
- Live 4-stage visual progress timeline.
- Real UPI QR Code gateway with advance vs. full payment options.
- Filterable directory of certified master craftsmen.

3. System Architecture & Technology Stack

GoldCare adopts a modern component-driven architecture designed for high maintainability, instantaneous navigation (HMR & Client-Side Routing), and maximum accessibility across devices.

Layer	Technology	Role & Key Justification
Frontend Framework	React.js 19 + Vite 8	Component lifecycle, lightning-fast dev bundling, and reactive state management.
Styling & Design System	Tailwind CSS v4 + Custom Tokens	Utility-first responsive layouts with tailored gold/cream palettes, glassmorphism & shimmer animations.
Routing Engine	React Router DOM v7 (HashRouter)	Enables seamless multi-page routing on GitHub Pages without requiring server-side 404 URL rewriting.
Icons & Assets	Lucide React	Scalable, crisp SVG iconography representing jewellery, security, and administrative actions.
Data Persistence	Browser LocalStorage API	Acts as an embedded client-side database with complete CRUD operations and fallback seed data.
Deployment & CI/CD	GitHub Actions + GitHub Pages	Automated linting, building, and static deployment triggered on every Git push to `main`.

Application Hierarchy & Routing Map

```
Root (HashRouter)
├── Public MainLayout (Navbar + Outlet + Footer)
│   ├── #/              ➔ Home Page (Hero, Trust Pillars, Services Carousel, Reviews)
│   ├── #/services      ➔ Service Catalog (10+ repairs with turnaround times & pricing)
│   ├── #/shops         ➔ Partner Shops Directory (Search, city filters, ratings)
│   ├── #/booking       ➔ 3-Step Interactive Booking Form
│   └── #/track         ➔ Live Repair Tracking & Payment Receipt Status
├── Standalone Route
│   └── #/admin/login   ➔ Secure Admin Authentication Portal (Protected Session)
└── Protected AdminLayout (Sidebar + ProtectedRoute + Admin Outlet)
    ├── #/admin/dashboard   ➔ Executive KPI Cards, Status Counters & Quick Actions
    ├── #/admin/shops       ➔ Shop CRUD (Add, Edit, Toggle Active/Inactive, Delete)
    ├── #/admin/bookings    ➔ Booking Table (Search, Status Filter, Remarks Slide-in Draw
    ├── #/admin/customers   ➔ Customer Directory with Aggregated History
```

```
    ├── #/admin/reports     ➔ Date-range Visual Analytics & CSV Data Export
    └── #/admin/settings    ➔ LocalStorage Diagnostics, Seed Reset & Quotas
```

4. Detailed Module Breakdown

4.1. The 3-Step Interactive Repair Booking Engine

The booking module (`src/pages/Booking.jsx`) allows customers to schedule their repairs seamlessly. The form includes strict client-side validation for:

- **Customer Info:** Full name, 10-digit Indian mobile number validation, and email.
- **Jewellery Details:** Selection from 11 jewellery classes (Gold Ring, Necklace, Bangles, Diamond Solitaire, etc.), repair category, and photo upload (max 2MB with live canvas preview).
- **Artisan Shop & Appointment:** Direct assignment to verified local partner workshops, appointment date restrictions (cannot select past dates), and preferred time slot selection.

4.2. Real UPI QR Code & Payment Escrow System

To ensure authentic business realism, GoldCare features a built-in Payment Modal (`src/components/PaymentModal.jsx`) that opens immediately after booking submission:

- **Dynamic UPI QR Generator:** Generates real scannable QR codes complying with the official NPCI UPI specifications: `upi://pay?pa=goldcare.jewels@okhdfcbank&pn=GoldCare%20Jewels&am=500&cu=INR&tn=GC-2026-0001`
- **Custom Merchant Insignia:** Features the GoldCare golden gem badge embedded directly in the center of the QR matrix, replicating merchant soundbox standees (Paytm/PhonePe).
- **5-Minute Live Countdown Timer:** Simulates gateway session timeout (`04:59` counting down per second).
- **Payment Modes:** Supports Advance Token (₹500), Full Estimated Amount, Debit/Credit Card, Net Banking (SBI, HDFC, ICICI, Axis), and Pay at Shop Counter.
- **Official GST Invoice:** Generates a printable Tax Receipt with a unique Transaction ID (e.g. `UPI/REF/984210`), printable directly using `window.print()`.

4.3. Real-Time Tracking & Pipeline Status

Customers can track any repair in real-time at `#/track` using their Booking ID and Mobile Number. The tracking dashboard displays:

- **Progress Pipeline:** Visual 4-step stepper (`Pending` ➔ `Confirmed` ➔ `In Progress` ➔ `Completed`).

- **Artisan Notes:** Live display of workshop remarks and estimated pickup times.
- **Payment Status:** Shows whether the booking is paid, advance settled, or due at counter, with a built-in "Pay Online" button for unpaid bookings.

4.4. Full-Featured Admin Back-Office

Secured by `ProtectedRoute.jsx`, the admin portal (`#/admin/login`) provides workshop administrators with complete operational control:

- **Shop Management:** Create, update, or deactivate repair centres across cities.
- **Booking Drawer:** Slide-in detail panel to update repair stages, record payments, and append customer remarks.
- **CSV Export & Analytics:** Filter reports by date and export full raw dataset to CSV via `exportCSV.js`.

5. Data Models & LocalStorage Schema

The system utilizes four primary keys within browser `localStorage`, abstracted through a robust service layer (`src/services/localStorageService.js`):

1. Bookings Schema (Key: `goldcare_bookings`)

```
{
  "id": "b-1729984920193",
  "bookingId": "GC-2026-0001",
  "customerName": "Aarav Patel",
  "mobile": "9876543210",
  "email": "aarav@example.com",
  "jewelleryType": "Gold Ring",
  "repairService": "Gold Ring Repair",
  "shopId": "shop-001",
  "shopName": "Raj Gold Works",
  "description": "Band resizing from size 12 to 14, high-gloss polish",
  "appointmentDate": "2026-11-05",
  "appointmentTime": "10:00",
  "budget": "1500",
  "status": "In Progress",              // Pending | Confirmed | In Progress | Completed
  "paymentStatus": "Paid",              // Pending | Paid | Pay at Counter
  "paidAmount": 500,
  "paymentMethod": "UPI QR Code",
  "transactionId": "UPI/REF/8849204",
  "adminRemarks": "Artisan has started resizing. Ready on Thursday.",
  "createdAt": "2026-10-26T17:30:00.000Z",
  "updatedAt": "2026-10-26T18:15:00.000Z"
}
```

2. Shop Schema (Key: `goldcare_shops`)

```
{
  "id": "shop-001",
  "name": "Raj Gold Works",
  "owner": "Rajesh Kumar",
  "contact": "9876543210",
  "city": "Mumbai",
  "address": "12, MG Road, Near Bus Stand",
  "services": ["Gold Ring Repair", "Polishing", "Stone Replacement"],
  "openTime": "09:00",
  "closeTime": "20:00",
```

```
  "status": "active",                    // active | inactive
  "rating": 4.8,
  "reviews": 312
}
```

6. Security, Validation & Business Logic

- **Client-Side Validation Engine (`src/utils/validation.js`):** Regex-based validation for 10-digit Indian phone numbers (`/^[6-9]\d{9}$/`), RFC email specifications, date boundaries (preventing past date selection), and image size enforcement (≤ 2MB).
- **Authentication Guard (`ProtectedRoute.jsx`):** Inspects `localStorage.getItem('goldcare_admin_session')` before rendering any admin layout. Unauthorized requests are automatically redirected to `#/admin/login` .
- **Zero Server Attack Surface:** Because there are no backend APIs, SQL databases, or server-side scripts, the application is inherently immune to SQL injection, remote code execution, or server denial-of-service vulnerabilities.

7. Testing & Quality Assurance

Test Scenario	Input / Action	Expected Result	Status
Form Validation	Submit booking with empty mobile & past date	Form displays error banners and scrolls to invalid input	Passed
UPI QR Generation	Submit valid repair form	Modal opens with live QR code, exact amount, and 5-min timer	Passed
Payment Persistence	Click "I Have Scanned & Paid"	Updates booking in LocalStorage; receipt renders with ref number	Passed
Repair Tracking	Search Booking ID + Mobile	Displays 4-step progress bar and matching payment badge	Passed
Admin Authorization	Direct access to `#/admin/dashboard` without login	Route intercepts and redirects to `#/admin/login`	Passed
Static Deployment	Deploy to GitHub Pages with `base: '/GoldCare/'`	Assets, routes, and styles load cleanly with 0 console errors	Passed

8. Future Scope & Enhancements

1. **Cloud Database Synchronization:** Integration with Firebase / Supabase for cross-device real-time sync across multiple physical stores.

2. **Automated WhatsApp Notifications:** Webhook triggers to send booking confirmations and pickup alerts directly to customer WhatsApp numbers.
3. **Computer Vision Gemstone Scanner:** Integration with camera APIs to analyze jewellery images and automatically estimate gold karatage and wear damage.

9. Conclusion

GoldCare demonstrates how modern frontend technologies (React 19, Tailwind CSS v4, and Vite) combined with thoughtful client-side state design can produce a sophisticated, commercially-viable web application without requiring expensive backend infrastructure. It provides a complete, polished solution for jewellery customers and craftsmen alike.